University of California, Santa Barbara

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Department of Computer Science   Santa Barbara, CA 93106-5110

August 23, 2026

Re: Letter of collaboration for proposal "PESOSE: Track 3: RedTwin: AI-Enabled Digital Twins for Continuous Red Teaming of Open-Source Ecosystems"

Dear NSF Program Directors:

If the proposal submitted by Dr. Giovanni Vigna and Dr. Christopher Kruegel entitled "PESOSE: Track 3: RedTwin: AI-Enabled Digital Twins for Continuous Red Teaming of Open-Source Ecosystems" is selected for funding by NSF, it is my intent to collaborate and/or commit resources as the director of the Early Research Scholars Program (ERSP) in the Department of Computer Science at UCSB.

ERSP is a year-long, dual-mentored research apprenticeship pairing small undergraduate teams with a faculty PI and a graduate mentor. RedTwin's core task of capturing a system's configuration, reconstructing it as a digital twin, and verifying that reconstruction is the kind of sustained, empirically grounded work to which an ERSP team, supported over a full academic year, is structured to contribute.

Professors Vigna and Kruegel have mentored ERSP teams for multiple years on problems drawn from their active research in the Systems Security Lab (aka SecLab). Their projects have not only provided undergraduate students with opportunities to experience cutting-edge research, but they have a demonstrated impact extending well beyond the year-long ERSP collaboration. For example, their project in 2023-2024 built "CVEX," a framework for dockerized, network-logged reproduction of individual CVE vulnerabilities. The framework continues to remain in active use in UCSB's SecLab; and Professors Vigna and Kruegel subsequently published CVE-GENIE (2025), an LLM-based framework for automated CVE reproduction. RedTwin's digital-twin construction extends this same reproduce-and-observe capability to full infrastructures.

Specifically, should RedTwin be funded, I will commit to placing a dedicated team of three to five second-year undergraduate researchers with Professor Vigna and Kruegel, co-advised by an assigned SecLab graduate mentor, to contribute directly to RedTwin's development.

Based on the scope described in the proposal, planned contributions include:

- curating and building candidate open-source infrastructure configurations to serve as case studies for the digital-twin pipeline, in the same spirit as the CVEX-records library the group's earlier ERSP team helped establish;
- assisting in the design and execution of test cases that exercise the proposed logic-based invariant checks and attack-graph preservation analysis used to verify twin representativeness;
- supporting the empirical evaluation of the vulnerability-chaining agents against known multi-step exploit scenarios;
- and contributing directly to the design of the public capture-the-flag competition the proposal describes, particularly challenges centered on constructing representative digital twins, an area in which the group's earlier ERSP-built CVEX website is directly relevant prior experience.

ERSP is a standing, annually renewed program with departmental support that predates this proposal and is not contingent on it. ERSP's collaboration with Dr. Vigna’s group is accordingly not limited to the award period. So long as Professors Vigna and Kruegel continue as ERSP mentors, ERSP will continue placing undergraduate teams on this line of work in future cohorts.

The proposal notes that open-source software underlies critical national infrastructure in health care, finance, utilities, and government. ERSP students work directly with practitioner-facing documentation and remediation, as in CVEX, whose stated purpose was making CVE records usable by the people responsible for acting on them. This experience is directly applicable to RedTwin's goal of producing assessments that are actionable for the maintainers and operators of the systems being modeled.

If funded, ERSP's participation as described here involves no subaward or exchange of funds between UC Santa Barbara's ERSP program and any other organization.

Please contact me with any questions regarding this letter or ERSP's role in the proposed work.

Sincerely,

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